

A disproof of Conjecture 00000000961

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Third cumulants

The conjecture assigns the coefficient $\sqrt{2}$ to the ratio of the third free cumulant and the third classical cumulant. At order three, however, the two moment–cumulant formulae coincide:

$$\kappa_3 = m_3 - 3m_2m_1 + 2m_1^3, \quad r_3 = m_3 - 3m_2m_1 + 2m_1^3.$$

The distinction between all set partitions and noncrossing partitions first affects the cumulant formula at order four, not order three.

Theorem 1. *The claimed universal ratio $\sqrt{2}$ is false.*

Proof. Let $X = -1$ with probability $2/3$ and $X = 2$ with probability $1/3$. Then

$$m_1 = 0, \quad m_2 = 2, \quad m_3 = 2.$$

Substitution gives $\kappa_3 = r_3 = 2$. Their ratio is therefore 1. Equivalently, the necessary identity $r_3^2 = 2\kappa_3^2$ would read $4 = 8$. $\square$

Lean 4 certificate

The project in `lean4/` checks the weighted moments as exact integer identities after multiplication by 3, implements both order-three formulas, and proves $4 \neq 8$. The certificate is core Lean and has no unproved placeholder or added axiom.